UK NATIONAL RAIL

PASSENGER BEHAVIOR, REVENUE TRENDS & OPERATIONAL PERFORMANCE

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GOAL

To evaluate passenger behavior, route performance, and revenue trends in order to identify inefficiencies and uncover opportunities for growth and operational improvement.

Focus Areas:

- Passenger purchasing behavior and ticket preferences
- Route demand, station performance, and peak travel patterns
- Revenue composition and the impact of delays and cancellations

BUSINESS OBJECTIVES:

Marketing, Operations, and Finance teams require clear visibility into customer behavior, service performance, and revenue drivers to guide strategic decisions.

Marketing

- How can we increase railcard adoption and customer engagement?
- What purchase behaviors should inform targeted campaigns?
- How can we leverage online purchasing trends to improve conversions?

Operations

- Which routes show high demand vs underutilization?
- When are peak travel periods, and how should resources be allocated?
- What are the primary drivers of delays and cancellations?

Sales & Revenue

- What is the revenue impact of delays and cancellations?
- How does ticket class distribution influence overall revenue performance?
- Which ticket types drive revenue stability and growth?

KEY INSIGHTS & BUSINESS IMPACT:

Low Railcard Utilization

66% of passengers **do not use** a railcard

→ Represents a significant opportunity to increase engagement and customer lifetime value

Revenue Leakage from Service Disruptions

32% of refunds are from cancellations

68% from delays

→ Improving reliability can directly reduce revenue loss

Price-Sensitive Customer Base

90% of tickets are **Standard Class**

→ Limited upsell to premium options highlights pricing sensitivity

Strong Digital Adoption

59% of purchases **are made online**

→ Enables targeted promotions and digital-first engagement strategies

KEY METRICS:

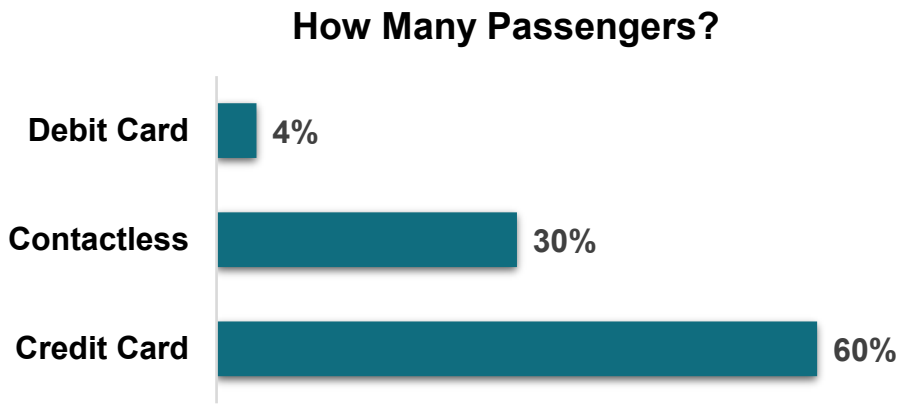
Snapshot of overall system performance and customer behavior

Total Passengers	% Standard Class	Net Revenue	Route Completion	Total Refund
31,653	90%	$703K	94%	($39k)

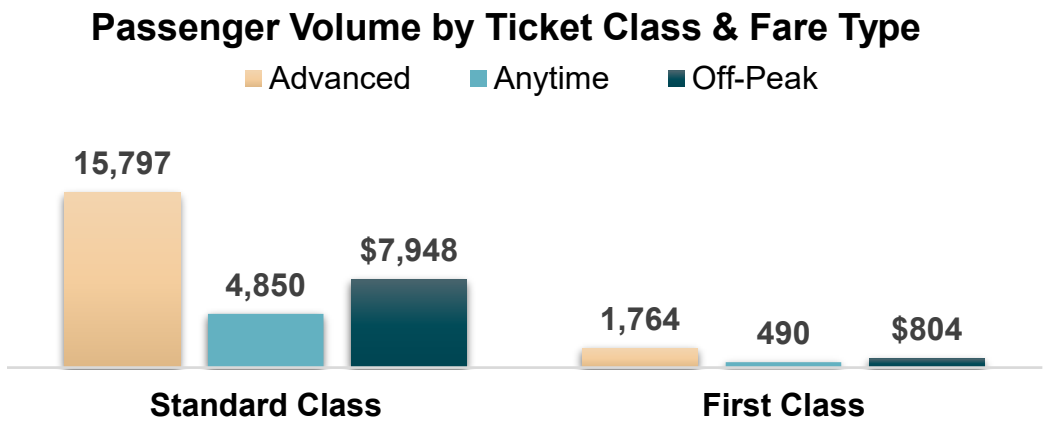
Increasing adoption by 10–15% could improve customer retention and perceived value without reducing base fares

CUSTOMER BEHAVIOR

Passenger purchasing behavior is highly digital and dominated by Standard Class ticket selection.



- 95% of transactions are card-based
- **65% credit card, 30% contactless**



- Standard Class drives **50% of total passenger volume**

Passenger behavior is highly digital and price-driven, reinforcing the importance of online engagement and competitive pricing strategies.

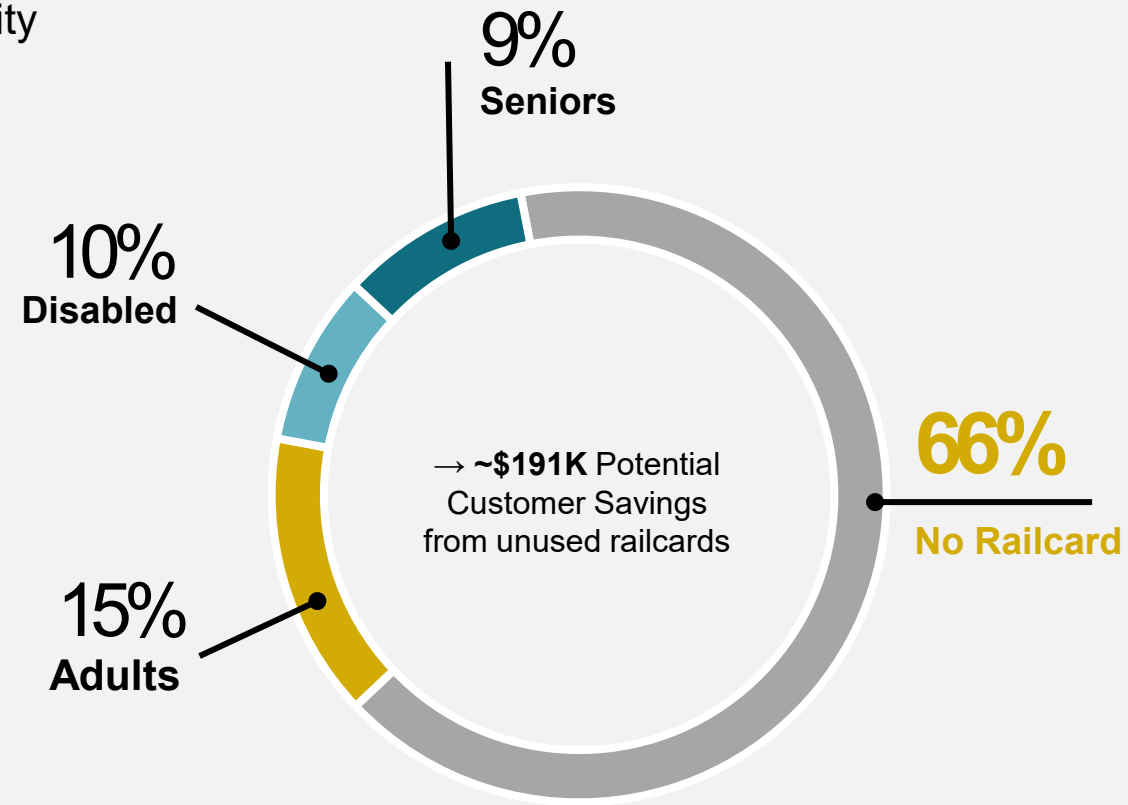
RAILCARD OVERVIEW

Low Railcard Adoption Is Driving Missed Revenue Opportunity

METRIC	VALUE
Unrealized Total Opportunity	$627K
10% Adoption	$63K
15% Adoption	$94K

- **~20,918** passengers did not use a railcard
- Represents **~$627K** in unrealized revenue
- **10–15% railcard adoption** increase = **$63K–$94K** incremental revenue

→ Assumes proportional adoption and average railcard price ($30)



Increasing railcard adoption presents a high-impact opportunity to improve both customer value and revenue, with potential gains of up to ~$94K.

ROUTE PERFORMANCE

Passenger demand is concentrated across key stations and peak travel periods.

DEPARTURE STATION	PASSENGERS	ON-TIME %	REVENUE
Liverpool Lime Street	4561	75%	**$135.27K**
London Euston	4954	**90%**	$112.05K
Manchester Piccadilly	**5650**	83%	$75.31K

ARRIVAL STATION	PASSENGERS	ON-TIME %	REVENUE
Birmingham New Street	**7742**	**91%**	$106.05K
Liverpool Lime Street	5022	87%	$33.76K
York	4019	**91%**	**$185.40K**

Peak travel occurs during **6–8 AM** and **5–7 PM**

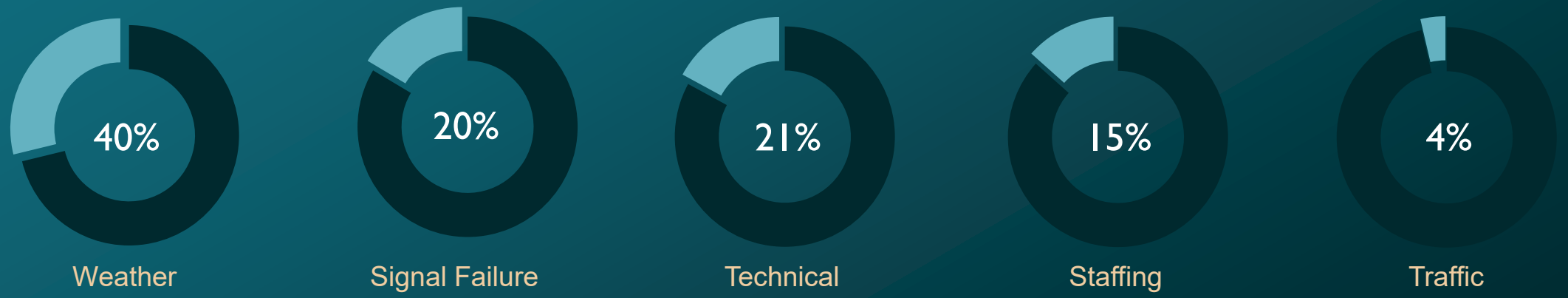
Manchester Piccadilly = busiest departure station **(18%)**

Birmingham New Street = busiest arrival station **(24%)**

Adjusting capacity and scheduling during peak periods can improve service efficiency and reduce congestion

OPERATIONAL IMPACT

Delay and Cancellation Drivers



Key Findings:

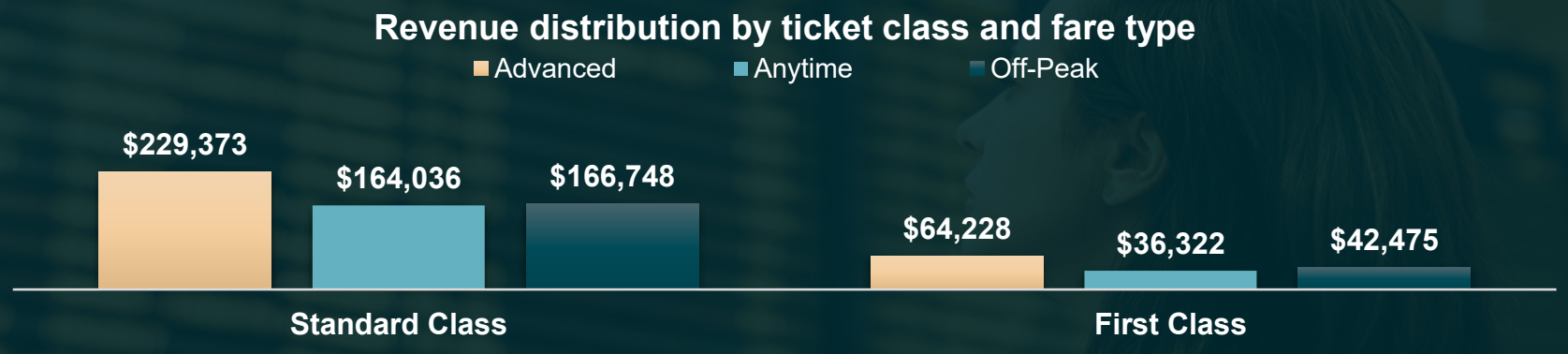
Technical issues are the leading cause of delays, accounting for **40% of disruptions**

Revenue Impact:

68% of refunds are due to delays
32% due to cancellations

Targeted operational improvements can reduce service disruptions and protect revenue

REVENUE DRIVERS & PRICING INSIGHTS



Standard Class Dominates Revenue

Standard Class drives the majority of ticket volume and revenue

→ Indicates strong price sensitivity among passengers

Advance Tickets Drive Volume

Advance Standard tickets represent a large share of purchases

→ Suggests customers prioritize lower pricing and early booking

Limited First Class Contribution

First Class generates only **~20%** of revenue

→ Highlights an opportunity to improve perceived value or target specific segments

Revenue is primarily driven by high-volume and price-sensitive ticket purchases.

THANK YOU

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Portfolio